

TELECOM APPLICATIONS DEVELOPMENT

Develops telecom software: billing, real-time charging, BSS/OSS, and VAS platforms. Covers protocols (SMPP, Diameter, SS7, SIGTRAN, SCTP), VOIP (Asterisk), and 4G/5G/IMS concepts. Built on Java, Python, C, and Erlang with REST, PostgreSQL/MongoDB, Linux, and Docker. A generative-AI strand adds LLM integration. Graduates build telecom apps, suited for Telecom Software, BSS/OSS, Backend, or VoIP Developer roles. This is a telecom software track with a specialized domain edge.

JOB PROFILES

**Telecom Software Developer · BSS-OSS Developer ·
VOIP Developer / Engineer (Asterisk/SIP) ·
Software Engineer (Telecom domain) · Linux System Administrator ·
Integration Engineer (Telecom / VAS) · Workflow Automation Specialist**



TOTAL HOURS

1,374 hrs

Theory + hands-on labs



DELIVERY

Blended

Online & on-site



UNIVERSITY GRADE

Good

Minimum grade required



CERTIFICATIONS

2


Where graduates work

Vodafone e& (Etisalat) Orange We Huawei Ericsson
Alcatel-Lucent VOIS Basata Edafa 4G Telecom VeriCash
Aman AWS Spectrum



Certifications earned

- RHCSA** — Red Hat Certified System Administrator
- CLF-C02** — AWS Certified Cloud Practitioner



Selection process

01 IQ & English Exam

- Online exam covering problem-solving aptitude and English proficiency.

02 Technical Exam

- Computer-based assessment covering the program's core technical fundamentals.

03 Technical Interview

- Interview with the technical committee.

04 Interpersonal Interview

- Candidates who pass the technical interview are invited to this final stage.



Who may apply

Open to graduates in Communications Engineering, Computer Engineering, and Computer Science with a minimum grade of Good or higher.

Program content – 1,374 hours

CATEGORY	HOURS	SHARE	KEY COURSES
Fundamental Courses	411 hrs	30%	C, CPP, HTML, CSS, JavaScript, Gen AI & Prompt Engineering, AI Workflow Automation, Gen-AI Development, AI Coding Agents, RAG and Agentic Software, Data Structures and Algorithms, Computer Networks, Agile, Software Engineering & UML
Core Courses	606 hrs	44%	2G, 3G, 4G, 5G, VAS, SMS, Broadband, Billing and Charging System, Mediation, Provisioning, eTOM, Java, Python, XML, JSON, Cloud, Docker, Git, REST, Android, Telecom Protocols, SS7 & IMS, Erlang/OTP, Fintech, VoIP Applications with Asterisk, IoT, Red Hat Admin I & II & III, Bash Shell, PostgreSQL, MongoDB
Soft Skills	57 hrs	4%	Communication Essentials for Professionals, High Impact Presentations, Job Seeking Skills, Professional Demeanor
Graduation Project	300 hrs	22%	

Hands-on labs are conducted in-person. Online components cover theory and self-study modules on MaharaTech.

Student Deliverable

Every graduate must obtain an international certification relevant to their specialization, confirming professional readiness for the field.

Before you apply

Explore these resources before your interview. Applicants are expected to have reviewed this material.

- Programming Fundamentals
<https://maharatech.gov.eg/course/view.php?id=2247>
- Object-Oriented Programming (OOP)
<https://maharatech.gov.eg/course/view.php?id=2206>
- Linux Fundamental
<https://maharatech.gov.eg/course/view.php?id=2155>
- Database Fundamentals
<https://maharatech.gov.eg/course/view.php?id=740>
- Mobile Networks Fundamentals
<https://www.tutorialspoint.com/gsm/index.htm>
- ITI Values & Mission
<https://iti.gov.eg/about-us>

APPLY AT

ITI.GOV.EG

Applications open each intake period. Check the website for current dates and deadlines.